

DESIGN FOR MANUFACTURABILITY (DFM) REPORT

Part: Part3 | Evaluation Date: 2026-08-16 20:25

Material: PP	Units: mm	Surface Texture: Textured	Draft Threshold: $\geq 3.0^{\circ}$
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1. Part Summary & Dimensions

Total Volume:	15062.33 mm ³	Bounding Box (X × Y × Z):	36.00 × 36.00 × 40.00 mm
Total Surface Area:	7562.06 mm ²	Topology Counts:	414 Faces, 957 Edges

2. Mold Pull Direction & 6-Axis Evaluation

Recommended Mold Opening Vector: (0.00, 0.00, 1.00) | Algorithm: 6-axis-cardinal-score | Confidence: HIGH

Candidate Axis	Undercut Faces	Zero Draft Faces	Recommendation
(+1, +0, +0)	187	31	Alternative
(-1, +0, +0)	187	31	Alternative
(+0, +1, +0)	165	75	Alternative
(+0, -1, +0)	165	75	Alternative
(+0, +0, +1)	37	94	RECOMMENDED
(+0, +0, -1)	274	94	Alternative

3. Draft Angle Analysis

Threshold Target: $\geq 3.0^{\circ}$ (Textured finish) | Pass: 274 | Low Draft: 0 | Zero Draft: 103 | Negative Draft: 37 (Total: 414 faces)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
0	+0.00°	(1.00, 0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
1	+90.00°	(0.00, 0.00, 1.00)	PASS ($\geq 3.0^{\circ}$)
2	+0.00°	(1.00, -0.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
3	-90.00°	(0.00, -0.00, -1.00)	NEGATIVE_DRAFT (BLOCKS EJECTION)
4	+0.00°	(1.00, -0.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
5	+90.00°	(0.00, 0.00, 1.00)	PASS ($\geq 3.0^{\circ}$)
6	+0.00°	(1.00, -0.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
7	-90.00°	(0.00, -0.00, -1.00)	NEGATIVE_DRAFT (BLOCKS EJECTION)
8	+4.21°	(1.00, -0.00, 0.07)	PASS ($\geq 3.0^{\circ}$)
9	-5.07°	(-1.00, 0.00, -0.09)	NEGATIVE_DRAFT (BLOCKS EJECTION)
10	+5.62°	(0.99, -0.00, 0.10)	PASS ($\geq 3.0^{\circ}$)
11	-5.62°	(0.99, -0.00, -0.10)	NEGATIVE_DRAFT (BLOCKS EJECTION)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
12	-5.62°	(0.99, -0.00, -0.10)	NEGATIVE_DRAFT (BLOCKS EJECTION)
13	+5.62°	(0.99, -0.00, 0.10)	PASS (≥ 3.0°)
14	-5.07°	(-1.00, 0.00, -0.09)	NEGATIVE_DRAFT (BLOCKS EJECTION)
15	+5.07°	(-1.00, 0.00, 0.09)	PASS (≥ 3.0°)
16	+5.07°	(-1.00, 0.00, 0.09)	PASS (≥ 3.0°)
17	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
18	+0.00°	(-1.00, 0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
19	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
20	+0.00°	(-1.00, 0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
21	-90.00°	(0.00, -0.00, -1.00)	NEGATIVE_DRAFT (BLOCKS EJECTION)
22	+0.00°	(-1.00, 0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
23	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
24	+0.00°	(-1.00, 0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
25	-90.00°	(0.00, -0.00, -1.00)	NEGATIVE_DRAFT (BLOCKS EJECTION)
26	+4.21°	(-1.00, 0.00, 0.07)	PASS (≥ 3.0°)
27	-5.07°	(1.00, -0.00, -0.09)	NEGATIVE_DRAFT (BLOCKS EJECTION)
28	+5.62°	(-0.99, 0.00, 0.10)	PASS (≥ 3.0°)
29	-5.62°	(-0.99, 0.00, -0.10)	NEGATIVE_DRAFT (BLOCKS EJECTION)
30	-5.62°	(-0.99, 0.00, -0.10)	NEGATIVE_DRAFT (BLOCKS EJECTION)
31	+5.62°	(-0.99, 0.00, 0.10)	PASS (≥ 3.0°)
32	-5.07°	(1.00, -0.00, -0.09)	NEGATIVE_DRAFT (BLOCKS EJECTION)
33	+5.07°	(1.00, -0.00, 0.09)	PASS (≥ 3.0°)
34	+5.07°	(1.00, -0.00, 0.09)	PASS (≥ 3.0°)
35	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
36	-90.00°	(0.00, -0.00, -1.00)	NEGATIVE_DRAFT (BLOCKS EJECTION)
37	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
38	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
39	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
40	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
41	+0.00°	(-0.77, 0.64, -0.00)	ZERO_DRAFT (VERTICAL WALL)
42	-0.00°	(-0.64, -0.77, -0.00)	ZERO_DRAFT (VERTICAL WALL)
43	+0.00°	(0.77, -0.64, -0.00)	ZERO_DRAFT (VERTICAL WALL)
44	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
45	+0.00°	(-0.34, 0.94, -0.00)	ZERO_DRAFT (VERTICAL WALL)
46	-0.00°	(-0.94, -0.34, -0.00)	ZERO_DRAFT (VERTICAL WALL)
47	+0.00°	(0.34, -0.94, -0.00)	ZERO_DRAFT (VERTICAL WALL)
48	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
49	+0.00°	(-0.94, -0.34, 0.00)	ZERO_DRAFT (VERTICAL WALL)
50	+0.00°	(0.34, -0.94, -0.00)	ZERO_DRAFT (VERTICAL WALL)
51	+0.00°	(0.94, 0.34, -0.00)	ZERO_DRAFT (VERTICAL WALL)
52	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
53	+0.00°	(-0.98, 0.17, -0.00)	ZERO_DRAFT (VERTICAL WALL)
54	-0.00°	(-0.17, -0.98, -0.00)	ZERO_DRAFT (VERTICAL WALL)
55	+0.00°	(0.98, -0.17, -0.00)	ZERO_DRAFT (VERTICAL WALL)
56	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
57	+0.00°	(0.77, -0.64, -0.00)	ZERO_DRAFT (VERTICAL WALL)
58	+0.00°	(0.64, 0.77, -0.00)	ZERO_DRAFT (VERTICAL WALL)
59	+0.00°	(-0.77, 0.64, -0.00)	ZERO_DRAFT (VERTICAL WALL)
60	+90.00°	(-0.00, 0.00, 1.00)	PASS (≥ 3.0°)
61	+0.00°	(0.34, -0.94, -0.00)	ZERO_DRAFT (VERTICAL WALL)
62	+0.00°	(0.94, 0.34, -0.00)	ZERO_DRAFT (VERTICAL WALL)
63	+0.00°	(-0.34, 0.94, -0.00)	ZERO_DRAFT (VERTICAL WALL)
64	+90.00°	(-0.00, 0.00, 1.00)	PASS (≥ 3.0°)
65	+0.00°	(-0.17, -0.98, 0.00)	ZERO_DRAFT (VERTICAL WALL)
66	+0.00°	(0.98, -0.17, -0.00)	ZERO_DRAFT (VERTICAL WALL)
67	+0.00°	(0.17, 0.98, -0.00)	ZERO_DRAFT (VERTICAL WALL)
68	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
69	+0.00°	(-0.64, -0.77, 0.00)	ZERO_DRAFT (VERTICAL WALL)
70	+0.00°	(0.77, -0.64, -0.00)	ZERO_DRAFT (VERTICAL WALL)
71	+0.00°	(0.64, 0.77, -0.00)	ZERO_DRAFT (VERTICAL WALL)
72	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
73	+0.00°	(0.94, 0.34, -0.00)	ZERO_DRAFT (VERTICAL WALL)
74	+0.00°	(-0.34, 0.94, 0.00)	ZERO_DRAFT (VERTICAL WALL)
75	+0.00°	(-0.94, -0.34, 0.00)	ZERO_DRAFT (VERTICAL WALL)
76	+90.00°	(0.00, -0.00, 1.00)	PASS (≥ 3.0°)
77	+0.00°	(0.98, -0.17, -0.00)	ZERO_DRAFT (VERTICAL WALL)
78	+0.00°	(0.17, 0.98, -0.00)	ZERO_DRAFT (VERTICAL WALL)
79	+0.00°	(-0.98, 0.17, -0.00)	ZERO_DRAFT (VERTICAL WALL)
80	+90.00°	(-0.00, 0.00, 1.00)	PASS (≥ 3.0°)
81	+0.00°	(0.64, 0.77, -0.00)	ZERO_DRAFT (VERTICAL WALL)
82	+0.00°	(-0.77, 0.64, -0.00)	ZERO_DRAFT (VERTICAL WALL)
83	+0.00°	(-0.64, -0.77, 0.00)	ZERO_DRAFT (VERTICAL WALL)
84	+90.00°	(0.00, -0.00, 1.00)	PASS (≥ 3.0°)
85	+0.00°	(0.17, 0.98, -0.00)	ZERO_DRAFT (VERTICAL WALL)
86	+0.00°	(-0.98, 0.17, -0.00)	ZERO_DRAFT (VERTICAL WALL)
87	+0.00°	(-0.17, -0.98, 0.00)	ZERO_DRAFT (VERTICAL WALL)
88	+90.00°	(0.00, -0.00, 1.00)	PASS (≥ 3.0°)
89	+0.00°	(-0.91, 0.42, 0.00)	ZERO_DRAFT (VERTICAL WALL)
90	+0.00°	(0.09, -1.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
91	+0.00°	(0.82, 0.57, -0.00)	ZERO_DRAFT (VERTICAL WALL)
92	+0.00°	(-0.57, 0.82, 0.00)	ZERO_DRAFT (VERTICAL WALL)
93	+0.00°	(0.57, -0.82, -0.00)	ZERO_DRAFT (VERTICAL WALL)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
94	+0.00°	(0.42, 0.91, -0.00)	ZERO_DRAFT (VERTICAL WALL)
95	+0.00°	(0.82, 0.57, -0.00)	ZERO_DRAFT (VERTICAL WALL)
96	-0.00°	(-1.00, -0.09, -0.00)	ZERO_DRAFT (VERTICAL WALL)
97	-0.00°	(-0.82, -0.57, -0.00)	ZERO_DRAFT (VERTICAL WALL)
98	+0.00°	(1.00, 0.09, -0.00)	ZERO_DRAFT (VERTICAL WALL)
99	+0.00°	(-0.57, 0.82, 0.00)	ZERO_DRAFT (VERTICAL WALL)
100	+0.00°	(1.00, 0.09, -0.00)	ZERO_DRAFT (VERTICAL WALL)
101	-0.00°	(-0.82, -0.57, -0.00)	ZERO_DRAFT (VERTICAL WALL)
102	-0.00°	(-0.42, -0.91, -0.00)	ZERO_DRAFT (VERTICAL WALL)
103	+0.00°	(-0.91, 0.42, 0.00)	ZERO_DRAFT (VERTICAL WALL)
104	+0.00°	(0.09, -1.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
105	+0.00°	(-0.09, 1.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
106	+0.00°	(0.42, 0.91, -0.00)	ZERO_DRAFT (VERTICAL WALL)
107	-0.00°	(-0.42, -0.91, -0.00)	ZERO_DRAFT (VERTICAL WALL)
108	+0.00°	(0.91, -0.42, -0.00)	ZERO_DRAFT (VERTICAL WALL)
109	+0.00°	(-0.09, 1.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
110	+0.00°	(0.91, -0.42, -0.00)	ZERO_DRAFT (VERTICAL WALL)
111	+0.00°	(-1.00, -0.09, 0.00)	ZERO_DRAFT (VERTICAL WALL)
112	+0.00°	(0.57, -0.82, -0.00)	ZERO_DRAFT (VERTICAL WALL)
113	+11.55°	(0.84, 0.51, 0.20)	PASS (≥ 3.0°)
114	+11.55°	(-0.98, -0.02, 0.20)	PASS (≥ 3.0°)
115	+11.55°	(0.98, 0.02, 0.20)	PASS (≥ 3.0°)
116	+11.55°	(-0.47, -0.86, 0.20)	PASS (≥ 3.0°)
117	+11.55°	(0.15, -0.97, 0.20)	PASS (≥ 3.0°)
118	+11.55°	(0.02, -0.98, 0.20)	PASS (≥ 3.0°)
119	+11.55°	(0.35, 0.91, 0.20)	PASS (≥ 3.0°)
120	+11.55°	(-0.91, 0.35, 0.20)	PASS (≥ 3.0°)
121	+11.55°	(0.76, 0.61, 0.20)	PASS (≥ 3.0°)
122	+11.55°	(-0.84, -0.51, 0.20)	PASS (≥ 3.0°)
123	+11.55°	(-0.51, 0.84, 0.20)	PASS (≥ 3.0°)
124	+11.55°	(-0.35, -0.91, 0.20)	PASS (≥ 3.0°)
125	+11.55°	(0.61, -0.76, 0.20)	PASS (≥ 3.0°)
126	+11.55°	(-0.97, -0.15, 0.20)	PASS (≥ 3.0°)
127	+11.55°	(0.91, -0.35, 0.20)	PASS (≥ 3.0°)
128	+11.55°	(0.47, 0.86, 0.20)	PASS (≥ 3.0°)
129	+11.55°	(0.86, -0.47, 0.20)	PASS (≥ 3.0°)
130	+11.55°	(-0.61, 0.76, 0.20)	PASS (≥ 3.0°)
131	+11.55°	(-0.15, 0.97, 0.20)	PASS (≥ 3.0°)
132	+11.55°	(-0.02, 0.98, 0.20)	PASS (≥ 3.0°)
133	+11.55°	(0.97, 0.15, 0.20)	PASS (≥ 3.0°)
134	+11.55°	(-0.76, -0.61, 0.20)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
135	+11.55°	(-0.86, 0.47, 0.20)	PASS (≥ 3.0°)
136	+11.55°	(0.51, -0.84, 0.20)	PASS (≥ 3.0°)
137	+64.83°	(-0.42, -0.03, 0.91)	PASS (≥ 3.0°)
138	+45.00°	(-0.46, -0.54, 0.71)	PASS (≥ 3.0°)
139	+54.38°	(-0.33, 0.48, 0.81)	PASS (≥ 3.0°)
140	+34.99°	(-0.82, -0.07, 0.57)	PASS (≥ 3.0°)
141	+64.81°	(0.17, 0.39, 0.91)	PASS (≥ 3.0°)
142	+45.00°	(-0.54, 0.46, 0.71)	PASS (≥ 3.0°)
143	+45.00°	(0.66, 0.24, 0.71)	PASS (≥ 3.0°)
144	+34.99°	(-0.07, 0.82, 0.57)	PASS (≥ 3.0°)
145	+34.99°	(0.35, 0.74, 0.57)	PASS (≥ 3.0°)
146	+45.00°	(0.46, 0.54, 0.71)	PASS (≥ 3.0°)
147	+45.00°	(-0.24, 0.66, 0.71)	PASS (≥ 3.0°)
148	+64.81°	(-0.05, 0.42, 0.91)	PASS (≥ 3.0°)
149	+34.99°	(-0.74, 0.35, 0.57)	PASS (≥ 3.0°)
150	+54.38°	(-0.53, 0.25, 0.81)	PASS (≥ 3.0°)
151	+45.00°	(-0.66, -0.24, 0.71)	PASS (≥ 3.0°)
152	+64.83°	(-0.35, -0.23, 0.91)	PASS (≥ 3.0°)
153	+64.83°	(-0.38, 0.19, 0.91)	PASS (≥ 3.0°)
154	+45.00°	(-0.12, -0.70, 0.71)	PASS (≥ 3.0°)
155	+54.38°	(-0.05, 0.58, 0.81)	PASS (≥ 3.0°)
156	+34.99°	(-0.67, -0.47, 0.57)	PASS (≥ 3.0°)
157	+64.81°	(0.34, 0.25, 0.91)	PASS (≥ 3.0°)
158	+45.00°	(-0.70, 0.12, 0.71)	PASS (≥ 3.0°)
159	+45.00°	(0.70, -0.12, 0.71)	PASS (≥ 3.0°)
160	+34.99°	(-0.47, 0.67, 0.57)	PASS (≥ 3.0°)
161	+34.99°	(0.67, 0.47, 0.57)	PASS (≥ 3.0°)
162	+45.00°	(0.12, 0.70, 0.71)	PASS (≥ 3.0°)
163	+45.00°	(0.12, 0.70, 0.71)	PASS (≥ 3.0°)
164	+64.81°	(-0.25, 0.34, 0.91)	PASS (≥ 3.0°)
165	+34.99°	(-0.47, 0.67, 0.57)	PASS (≥ 3.0°)
166	+54.38°	(-0.58, -0.05, 0.81)	PASS (≥ 3.0°)
167	+45.00°	(-0.70, 0.12, 0.71)	PASS (≥ 3.0°)
168	+64.83°	(-0.19, -0.38, 0.91)	PASS (≥ 3.0°)
169	+64.83°	(-0.23, 0.35, 0.91)	PASS (≥ 3.0°)
170	+45.00°	(0.24, -0.66, 0.71)	PASS (≥ 3.0°)
171	+54.38°	(0.25, 0.53, 0.81)	PASS (≥ 3.0°)
172	+34.99°	(-0.35, -0.74, 0.57)	PASS (≥ 3.0°)
173	+64.81°	(0.42, 0.05, 0.91)	PASS (≥ 3.0°)
174	+45.00°	(-0.66, -0.24, 0.71)	PASS (≥ 3.0°)
175	+45.00°	(0.54, -0.46, 0.71)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
176	+34.99°	(-0.74, 0.35, 0.57)	PASS (≥ 3.0°)
177	+34.99°	(0.82, 0.07, 0.57)	PASS (≥ 3.0°)
178	+45.00°	(-0.24, 0.66, 0.71)	PASS (≥ 3.0°)
179	+45.00°	(0.46, 0.54, 0.71)	PASS (≥ 3.0°)
180	+64.81°	(-0.39, 0.17, 0.91)	PASS (≥ 3.0°)
181	+34.99°	(-0.07, 0.82, 0.57)	PASS (≥ 3.0°)
182	+54.38°	(-0.48, -0.33, 0.81)	PASS (≥ 3.0°)
183	+45.00°	(-0.54, 0.46, 0.71)	PASS (≥ 3.0°)
184	+64.83°	(0.03, -0.42, 0.91)	PASS (≥ 3.0°)
185	+64.83°	(-0.03, 0.42, 0.91)	PASS (≥ 3.0°)
186	+45.00°	(0.54, -0.46, 0.71)	PASS (≥ 3.0°)
187	+54.38°	(0.48, 0.33, 0.81)	PASS (≥ 3.0°)
188	+34.99°	(0.07, -0.82, 0.57)	PASS (≥ 3.0°)
189	+64.81°	(0.39, -0.17, 0.91)	PASS (≥ 3.0°)
190	+45.00°	(-0.46, -0.54, 0.71)	PASS (≥ 3.0°)
191	+45.00°	(0.24, -0.66, 0.71)	PASS (≥ 3.0°)
192	+34.99°	(-0.82, -0.07, 0.57)	PASS (≥ 3.0°)
193	+34.99°	(0.74, -0.35, 0.57)	PASS (≥ 3.0°)
194	+45.00°	(-0.54, 0.46, 0.71)	PASS (≥ 3.0°)
195	+45.00°	(0.66, 0.24, 0.71)	PASS (≥ 3.0°)
196	+64.81°	(-0.42, -0.05, 0.91)	PASS (≥ 3.0°)
197	+34.99°	(0.35, 0.74, 0.57)	PASS (≥ 3.0°)
198	+54.38°	(-0.25, -0.53, 0.81)	PASS (≥ 3.0°)
199	+45.00°	(-0.24, 0.66, 0.71)	PASS (≥ 3.0°)
200	+64.83°	(0.23, -0.35, 0.91)	PASS (≥ 3.0°)
201	+64.83°	(0.19, 0.38, 0.91)	PASS (≥ 3.0°)
202	+45.00°	(0.70, -0.12, 0.71)	PASS (≥ 3.0°)
203	+54.38°	(0.58, 0.05, 0.81)	PASS (≥ 3.0°)
204	+34.99°	(0.47, -0.67, 0.57)	PASS (≥ 3.0°)
205	+64.81°	(0.25, -0.34, 0.91)	PASS (≥ 3.0°)
206	+45.00°	(-0.12, -0.70, 0.71)	PASS (≥ 3.0°)
207	+45.00°	(-0.12, -0.70, 0.71)	PASS (≥ 3.0°)
208	+34.99°	(-0.67, -0.47, 0.57)	PASS (≥ 3.0°)
209	+34.99°	(0.47, -0.67, 0.57)	PASS (≥ 3.0°)
210	+45.00°	(-0.70, 0.12, 0.71)	PASS (≥ 3.0°)
211	+45.00°	(0.70, -0.12, 0.71)	PASS (≥ 3.0°)
212	+64.81°	(-0.34, -0.25, 0.91)	PASS (≥ 3.0°)
213	+34.99°	(0.67, 0.47, 0.57)	PASS (≥ 3.0°)
214	+54.38°	(0.05, -0.58, 0.81)	PASS (≥ 3.0°)
215	+45.00°	(0.12, 0.70, 0.71)	PASS (≥ 3.0°)
216	+64.83°	(0.38, -0.19, 0.91)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
217	+64.83°	(0.35, 0.23, 0.91)	PASS (≥ 3.0°)
218	+45.00°	(0.66, 0.24, 0.71)	PASS (≥ 3.0°)
219	+54.38°	(0.53, -0.25, 0.81)	PASS (≥ 3.0°)
220	+34.99°	(0.74, -0.35, 0.57)	PASS (≥ 3.0°)
221	+64.81°	(0.05, -0.42, 0.91)	PASS (≥ 3.0°)
222	+45.00°	(0.24, -0.66, 0.71)	PASS (≥ 3.0°)
223	+45.00°	(-0.46, -0.54, 0.71)	PASS (≥ 3.0°)
224	+34.99°	(-0.35, -0.74, 0.57)	PASS (≥ 3.0°)
225	+34.99°	(0.07, -0.82, 0.57)	PASS (≥ 3.0°)
226	+45.00°	(-0.66, -0.24, 0.71)	PASS (≥ 3.0°)
227	+45.00°	(0.54, -0.46, 0.71)	PASS (≥ 3.0°)
228	+64.81°	(-0.17, -0.39, 0.91)	PASS (≥ 3.0°)
229	+34.99°	(0.82, 0.07, 0.57)	PASS (≥ 3.0°)
230	+54.38°	(0.33, -0.48, 0.81)	PASS (≥ 3.0°)
231	+45.00°	(0.46, 0.54, 0.71)	PASS (≥ 3.0°)
232	+64.83°	(0.42, 0.03, 0.91)	PASS (≥ 3.0°)
233	+37.89°	(0.23, 0.75, 0.61)	PASS (≥ 3.0°)
234	+45.00°	(-0.24, 0.66, 0.71)	PASS (≥ 3.0°)
235	+56.07°	(0.24, 0.51, 0.83)	PASS (≥ 3.0°)
236	+45.00°	(0.66, 0.24, 0.71)	PASS (≥ 3.0°)
237	+56.07°	(0.51, -0.24, 0.83)	PASS (≥ 3.0°)
238	+45.00°	(0.24, -0.66, 0.71)	PASS (≥ 3.0°)
239	+37.89°	(0.66, -0.43, 0.61)	PASS (≥ 3.0°)
240	+37.89°	(-0.17, 0.77, 0.61)	PASS (≥ 3.0°)
241	+45.00°	(-0.54, 0.46, 0.71)	PASS (≥ 3.0°)
242	+56.07°	(-0.05, 0.56, 0.83)	PASS (≥ 3.0°)
243	+45.00°	(0.46, 0.54, 0.71)	PASS (≥ 3.0°)
244	+56.07°	(0.56, 0.05, 0.83)	PASS (≥ 3.0°)
245	+45.00°	(0.54, -0.46, 0.71)	PASS (≥ 3.0°)
246	+37.89°	(0.79, -0.04, 0.61)	PASS (≥ 3.0°)
247	+37.89°	(-0.75, 0.23, 0.61)	PASS (≥ 3.0°)
248	+45.00°	(-0.66, -0.24, 0.71)	PASS (≥ 3.0°)
249	+56.07°	(-0.51, 0.24, 0.83)	PASS (≥ 3.0°)
250	+45.00°	(-0.24, 0.66, 0.71)	PASS (≥ 3.0°)
251	+56.07°	(0.24, 0.51, 0.83)	PASS (≥ 3.0°)
252	+45.00°	(0.66, 0.24, 0.71)	PASS (≥ 3.0°)
253	+37.89°	(0.43, 0.66, 0.61)	PASS (≥ 3.0°)
254	+37.89°	(-0.58, -0.54, 0.61)	PASS (≥ 3.0°)
255	+45.00°	(-0.12, -0.70, 0.71)	PASS (≥ 3.0°)
256	+56.07°	(-0.46, -0.32, 0.83)	PASS (≥ 3.0°)
257	+45.00°	(-0.70, 0.12, 0.71)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
258	+56.07°	(-0.32, 0.46, 0.83)	PASS (≥ 3.0°)
259	+45.00°	(0.12, 0.70, 0.71)	PASS (≥ 3.0°)
260	+37.89°	(-0.36, 0.70, 0.61)	PASS (≥ 3.0°)
261	+37.89°	(0.75, -0.23, 0.61)	PASS (≥ 3.0°)
262	+45.00°	(0.66, 0.24, 0.71)	PASS (≥ 3.0°)
263	+56.07°	(0.51, -0.24, 0.83)	PASS (≥ 3.0°)
264	+45.00°	(0.24, -0.66, 0.71)	PASS (≥ 3.0°)
265	+56.07°	(-0.24, -0.51, 0.83)	PASS (≥ 3.0°)
266	+45.00°	(-0.66, -0.24, 0.71)	PASS (≥ 3.0°)
267	+37.89°	(-0.43, -0.66, 0.61)	PASS (≥ 3.0°)
268	+37.89°	(0.54, -0.58, 0.61)	PASS (≥ 3.0°)
269	+45.00°	(0.70, -0.12, 0.71)	PASS (≥ 3.0°)
270	+56.07°	(0.32, -0.46, 0.83)	PASS (≥ 3.0°)
271	+45.00°	(-0.12, -0.70, 0.71)	PASS (≥ 3.0°)
272	+56.07°	(-0.46, -0.32, 0.83)	PASS (≥ 3.0°)
273	+45.00°	(-0.70, 0.12, 0.71)	PASS (≥ 3.0°)
274	+37.89°	(-0.70, -0.36, 0.61)	PASS (≥ 3.0°)
275	+37.89°	(0.58, 0.54, 0.61)	PASS (≥ 3.0°)
276	+45.00°	(0.12, 0.70, 0.71)	PASS (≥ 3.0°)
277	+56.07°	(0.46, 0.32, 0.83)	PASS (≥ 3.0°)
278	+45.00°	(0.70, -0.12, 0.71)	PASS (≥ 3.0°)
279	+56.07°	(0.32, -0.46, 0.83)	PASS (≥ 3.0°)
280	+45.00°	(-0.12, -0.70, 0.71)	PASS (≥ 3.0°)
281	+37.89°	(0.36, -0.70, 0.61)	PASS (≥ 3.0°)
282	+37.89°	(0.17, -0.77, 0.61)	PASS (≥ 3.0°)
283	+45.00°	(0.54, -0.46, 0.71)	PASS (≥ 3.0°)
284	+56.07°	(0.05, -0.56, 0.83)	PASS (≥ 3.0°)
285	+45.00°	(-0.46, -0.54, 0.71)	PASS (≥ 3.0°)
286	+56.07°	(-0.56, -0.05, 0.83)	PASS (≥ 3.0°)
287	+45.00°	(-0.54, 0.46, 0.71)	PASS (≥ 3.0°)
288	+37.89°	(-0.79, 0.04, 0.61)	PASS (≥ 3.0°)
289	+37.89°	(-0.23, -0.75, 0.61)	PASS (≥ 3.0°)
290	+45.00°	(0.24, -0.66, 0.71)	PASS (≥ 3.0°)
291	+56.07°	(-0.24, -0.51, 0.83)	PASS (≥ 3.0°)
292	+45.00°	(-0.66, -0.24, 0.71)	PASS (≥ 3.0°)
293	+56.07°	(-0.51, 0.24, 0.83)	PASS (≥ 3.0°)
294	+45.00°	(-0.24, 0.66, 0.71)	PASS (≥ 3.0°)
295	+37.89°	(-0.66, 0.43, 0.61)	PASS (≥ 3.0°)
296	+37.89°	(-0.77, -0.17, 0.61)	PASS (≥ 3.0°)
297	+45.00°	(-0.46, -0.54, 0.71)	PASS (≥ 3.0°)
298	+56.07°	(-0.56, -0.05, 0.83)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
299	+45.00°	(-0.54, 0.46, 0.71)	PASS (≥ 3.0°)
300	+56.07°	(-0.05, 0.56, 0.83)	PASS (≥ 3.0°)
301	+45.00°	(0.46, 0.54, 0.71)	PASS (≥ 3.0°)
302	+37.89°	(0.04, 0.79, 0.61)	PASS (≥ 3.0°)
303	+37.89°	(-0.54, 0.58, 0.61)	PASS (≥ 3.0°)
304	+45.00°	(-0.70, 0.12, 0.71)	PASS (≥ 3.0°)
305	+56.07°	(-0.32, 0.46, 0.83)	PASS (≥ 3.0°)
306	+45.00°	(0.12, 0.70, 0.71)	PASS (≥ 3.0°)
307	+56.07°	(0.46, 0.32, 0.83)	PASS (≥ 3.0°)
308	+45.00°	(0.70, -0.12, 0.71)	PASS (≥ 3.0°)
309	+37.89°	(0.70, 0.36, 0.61)	PASS (≥ 3.0°)
310	+37.89°	(0.77, 0.17, 0.61)	PASS (≥ 3.0°)
311	+45.00°	(0.46, 0.54, 0.71)	PASS (≥ 3.0°)
312	+56.07°	(0.56, 0.05, 0.83)	PASS (≥ 3.0°)
313	+45.00°	(0.54, -0.46, 0.71)	PASS (≥ 3.0°)
314	+56.07°	(0.05, -0.56, 0.83)	PASS (≥ 3.0°)
315	+45.00°	(-0.46, -0.54, 0.71)	PASS (≥ 3.0°)
316	+37.89°	(-0.04, -0.79, 0.61)	PASS (≥ 3.0°)
317	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
318	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
319	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
320	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
321	+0.00°	N/A	ZERO_DRAFT (VERTICAL WALL)
322	+0.00°	(-1.00, 0.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
323	+0.00°	(1.00, -0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
324	+0.00°	(0.00, 1.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
325	+90.00°	(0.00, 0.00, 1.00)	PASS (≥ 3.0°)
326	+45.00°	(0.00, 0.71, 0.71)	PASS (≥ 3.0°)
327	+48.34°	(0.67, -0.00, 0.75)	PASS (≥ 3.0°)
328	+35.11°	(0.60, 0.55, 0.57)	PASS (≥ 3.0°)
329	+0.00°	(0.75, 0.67, -0.00)	ZERO_DRAFT (VERTICAL WALL)
330	+29.85°	(0.72, 0.48, 0.50)	PASS (≥ 3.0°)
331	+45.00°	(0.71, -0.00, 0.71)	PASS (≥ 3.0°)
332	+35.07°	(0.61, 0.54, 0.57)	PASS (≥ 3.0°)
333	+0.00°	(0.76, 0.65, -0.00)	ZERO_DRAFT (VERTICAL WALL)
334	-35.07°	(0.61, 0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
335	-45.00°	(0.71, -0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
336	-29.86°	(0.72, 0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
337	+0.00°	(0.75, 0.67, -0.00)	ZERO_DRAFT (VERTICAL WALL)
338	+29.85°	(0.72, 0.48, 0.50)	PASS (≥ 3.0°)
339	+45.00°	(0.71, -0.00, 0.71)	PASS (≥ 3.0°)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
340	+35.07°	(0.61, 0.54, 0.57)	PASS ($\geq 3.0^\circ$)
341	+0.00°	(0.76, 0.65, -0.00)	ZERO_DRAFT (VERTICAL WALL)
342	-35.07°	(0.61, 0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
343	-45.00°	(0.71, -0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
344	-29.85°	(0.72, 0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
345	+48.34°	(-0.67, 0.00, 0.75)	PASS ($\geq 3.0^\circ$)
346	+35.11°	(-0.60, 0.55, 0.57)	PASS ($\geq 3.0^\circ$)
347	+0.00°	(-0.75, 0.67, 0.00)	ZERO_DRAFT (VERTICAL WALL)
348	+29.86°	(-0.72, 0.48, 0.50)	PASS ($\geq 3.0^\circ$)
349	+45.00°	(-0.71, 0.00, 0.71)	PASS ($\geq 3.0^\circ$)
350	+35.07°	(-0.61, 0.54, 0.57)	PASS ($\geq 3.0^\circ$)
351	+0.00°	(-0.76, 0.65, 0.00)	ZERO_DRAFT (VERTICAL WALL)
352	-35.07°	(-0.61, 0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
353	-45.00°	(-0.71, 0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
354	-29.85°	(-0.72, 0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
355	+0.00°	(-0.75, 0.67, 0.00)	ZERO_DRAFT (VERTICAL WALL)
356	+29.86°	(-0.72, 0.48, 0.50)	PASS ($\geq 3.0^\circ$)
357	+45.00°	(-0.71, 0.00, 0.71)	PASS ($\geq 3.0^\circ$)
358	+35.07°	(-0.61, 0.54, 0.57)	PASS ($\geq 3.0^\circ$)
359	+0.00°	(-0.76, 0.65, 0.00)	ZERO_DRAFT (VERTICAL WALL)
360	-35.07°	(-0.61, 0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
361	-45.00°	(-0.71, 0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
362	-29.85°	(-0.72, 0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
363	-0.00°	(-1.00, -0.00, -0.00)	ZERO_DRAFT (VERTICAL WALL)
364	+0.00°	(1.00, 0.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
365	+0.00°	(0.00, -1.00, 0.00)	ZERO_DRAFT (VERTICAL WALL)
366	+90.00°	(0.00, 0.00, 1.00)	PASS ($\geq 3.0^\circ$)
367	+45.00°	(0.00, -0.71, 0.71)	PASS ($\geq 3.0^\circ$)
368	+48.34°	(0.67, 0.00, 0.75)	PASS ($\geq 3.0^\circ$)
369	+35.11°	(0.60, -0.55, 0.57)	PASS ($\geq 3.0^\circ$)
370	+0.00°	(0.75, -0.67, -0.00)	ZERO_DRAFT (VERTICAL WALL)
371	+29.86°	(0.72, -0.48, 0.50)	PASS ($\geq 3.0^\circ$)
372	+45.00°	(0.71, 0.00, 0.71)	PASS ($\geq 3.0^\circ$)
373	+35.07°	(0.61, -0.54, 0.57)	PASS ($\geq 3.0^\circ$)
374	+0.00°	(0.76, -0.65, -0.00)	ZERO_DRAFT (VERTICAL WALL)
375	-35.07°	(0.61, -0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
376	-45.00°	(0.71, 0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
377	-29.85°	(0.72, -0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
378	+0.00°	(0.75, -0.67, -0.00)	ZERO_DRAFT (VERTICAL WALL)
379	+29.86°	(0.72, -0.48, 0.50)	PASS ($\geq 3.0^\circ$)
380	+45.00°	(0.71, 0.00, 0.71)	PASS ($\geq 3.0^\circ$)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
381	+35.07°	(0.61, -0.54, 0.57)	PASS (≥ 3.0°)
382	+0.00°	(0.76, -0.65, -0.00)	ZERO_DRAFT (VERTICAL WALL)
383	-35.07°	(0.61, -0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
384	-45.00°	(0.71, 0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
385	-29.85°	(0.72, -0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
386	+48.34°	(-0.67, -0.00, 0.75)	PASS (≥ 3.0°)
387	+35.11°	(-0.60, -0.55, 0.57)	PASS (≥ 3.0°)
388	-0.00°	(-0.75, -0.67, -0.00)	ZERO_DRAFT (VERTICAL WALL)
389	+29.85°	(-0.72, -0.48, 0.50)	PASS (≥ 3.0°)
390	+45.00°	(-0.71, -0.00, 0.71)	PASS (≥ 3.0°)
391	+35.07°	(-0.61, -0.54, 0.57)	PASS (≥ 3.0°)
392	-0.00°	(-0.76, -0.65, -0.00)	ZERO_DRAFT (VERTICAL WALL)
393	-35.07°	(-0.61, -0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
394	-45.00°	(-0.71, -0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
395	-29.86°	(-0.72, -0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
396	-0.00°	(-0.75, -0.67, -0.00)	ZERO_DRAFT (VERTICAL WALL)
397	+29.85°	(-0.72, -0.48, 0.50)	PASS (≥ 3.0°)
398	+45.00°	(-0.71, -0.00, 0.71)	PASS (≥ 3.0°)
399	+35.07°	(-0.61, -0.54, 0.57)	PASS (≥ 3.0°)
400	-0.00°	(-0.76, -0.65, -0.00)	ZERO_DRAFT (VERTICAL WALL)
401	-35.07°	(-0.61, -0.54, -0.57)	NEGATIVE_DRAFT (BLOCKS EJECTION)
402	-45.00°	(-0.71, -0.00, -0.71)	NEGATIVE_DRAFT (BLOCKS EJECTION)
403	-29.85°	(-0.72, -0.48, -0.50)	NEGATIVE_DRAFT (BLOCKS EJECTION)
404	+54.45°	(0.00, -0.58, 0.81)	PASS (≥ 3.0°)
405	+54.45°	(0.00, 0.58, 0.81)	PASS (≥ 3.0°)
406	+0.00°	(0.71, -0.70, 0.00)	ZERO_DRAFT (VERTICAL WALL)
407	+52.80°	(0.43, -0.43, 0.80)	PASS (≥ 3.0°)
408	+0.00°	(-0.71, -0.70, 0.00)	ZERO_DRAFT (VERTICAL WALL)
409	+52.80°	(-0.43, -0.43, 0.80)	PASS (≥ 3.0°)
410	+0.00°	(0.71, 0.70, -0.00)	ZERO_DRAFT (VERTICAL WALL)
411	+52.80°	(0.43, 0.43, 0.80)	PASS (≥ 3.0°)
412	+0.00°	(-0.71, 0.70, 0.00)	ZERO_DRAFT (VERTICAL WALL)
413	+52.80°	(-0.43, 0.43, 0.80)	PASS (≥ 3.0°)

4. Undercuts & Resolution Plan

Detected 48 undercut face(s) creating mechanical mold locks. Review the automated resolution plan below:

Face ID	Resolution	Action Vector	Engineering Rationale & Tooling Requirement
330	Side-Action (Slider)	(0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, 0.56, 0.00) (+X); requires cam/hydraulic slide.

Face ID	Resolution	Action Vector	Engineering Rationale & Tooling Requirement
331	Side-Action (Slider)	(1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, -0.00, 0.00) (+X); requires cam/hydraulic slide.
332	Side-Action (Slider)	(0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, 0.66, 0.00) (+X); requires cam/hydraulic slide.
334	Side-Action (Slider)	(0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, 0.66, 0.00) (+X); requires cam/hydraulic slide.
335	Side-Action (Slider)	(1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, -0.00, 0.00) (+X); requires cam/hydraulic slide.
336	Side-Action (Slider)	(0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, 0.56, 0.00) (+X); requires cam/hydraulic slide.
338	Side-Action (Slider)	(0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, 0.56, 0.00) (+X); requires cam/hydraulic slide.
339	Side-Action (Slider)	(1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, -0.00, 0.00) (+X); requires cam/hydraulic slide.
340	Side-Action (Slider)	(0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, 0.66, 0.00) (+X); requires cam/hydraulic slide.
342	Side-Action (Slider)	(0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, 0.66, 0.00) (+X); requires cam/hydraulic slide.
343	Side-Action (Slider)	(1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, -0.00, 0.00) (+X); requires cam/hydraulic slide.
344	Side-Action (Slider)	(0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, 0.56, 0.00) (+X); requires cam/hydraulic slide.
348	Side-Action (Slider)	(-0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, 0.56, 0.00) (-X); requires cam/hydraulic slide.
349	Side-Action (Slider)	(-1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, 0.00, 0.00) (-X); requires cam/hydraulic slide.
350	Side-Action (Slider)	(-0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, 0.66, 0.00) (-X); requires cam/hydraulic slide.
352	Side-Action (Slider)	(-0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, 0.66, 0.00) (-X); requires cam/hydraulic slide.
353	Side-Action (Slider)	(-1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, 0.00, 0.00) (-X); requires cam/hydraulic slide.
354	Side-Action (Slider)	(-0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, 0.56, 0.00) (-X); requires cam/hydraulic slide.
356	Side-Action (Slider)	(-0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, 0.56, 0.00) (-X); requires cam/hydraulic slide.
357	Side-Action (Slider)	(-1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, 0.00, 0.00) (-X); requires cam/hydraulic slide.
358	Side-Action (Slider)	(-0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, 0.66, 0.00) (-X); requires cam/hydraulic slide.
360	Side-Action (Slider)	(-0.75, 0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, 0.66, 0.00) (-X); requires cam/hydraulic slide.
361	Side-Action (Slider)	(-1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, 0.00, 0.00) (-X); requires cam/hydraulic slide.
362	Side-Action (Slider)	(-0.83, 0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, 0.56, 0.00) (-X); requires cam/hydraulic slide.

Face ID	Resolution	Action Vector	Engineering Rationale & Tooling Requirement
371	Side-Action (Slider)	(0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, -0.56, 0.00) (+X); requires cam/hydraulic slide.
372	Side-Action (Slider)	(1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, 0.00, 0.00) (+X); requires cam/hydraulic slide.
373	Side-Action (Slider)	(0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, -0.66, 0.00) (+X); requires cam/hydraulic slide.
375	Side-Action (Slider)	(0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, -0.66, 0.00) (+X); requires cam/hydraulic slide.
376	Side-Action (Slider)	(1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, 0.00, 0.00) (+X); requires cam/hydraulic slide.
377	Side-Action (Slider)	(0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, -0.56, 0.00) (+X); requires cam/hydraulic slide.
379	Side-Action (Slider)	(0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, -0.56, 0.00) (+X); requires cam/hydraulic slide.
380	Side-Action (Slider)	(1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, 0.00, 0.00) (+X); requires cam/hydraulic slide.
381	Side-Action (Slider)	(0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, -0.66, 0.00) (+X); requires cam/hydraulic slide.
383	Side-Action (Slider)	(0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (0.75, -0.66, 0.00) (+X); requires cam/hydraulic slide.
384	Side-Action (Slider)	(1.00, 0.00, 0.00)	External lateral undercut with clear side-draw access along (1.00, 0.00, 0.00) (+X); requires cam/hydraulic slide.
385	Side-Action (Slider)	(0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (0.83, -0.56, 0.00) (+X); requires cam/hydraulic slide.
389	Side-Action (Slider)	(-0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, -0.56, 0.00) (-X); requires cam/hydraulic slide.
390	Side-Action (Slider)	(-1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, -0.00, 0.00) (-X); requires cam/hydraulic slide.
391	Side-Action (Slider)	(-0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, -0.66, 0.00) (-X); requires cam/hydraulic slide.
393	Side-Action (Slider)	(-0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, -0.66, 0.00) (-X); requires cam/hydraulic slide.
394	Side-Action (Slider)	(-1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, -0.00, 0.00) (-X); requires cam/hydraulic slide.
395	Side-Action (Slider)	(-0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, -0.56, 0.00) (-X); requires cam/hydraulic slide.
397	Side-Action (Slider)	(-0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, -0.56, 0.00) (-X); requires cam/hydraulic slide.
398	Side-Action (Slider)	(-1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, -0.00, 0.00) (-X); requires cam/hydraulic slide.
399	Side-Action (Slider)	(-0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, -0.66, 0.00) (-X); requires cam/hydraulic slide.
401	Side-Action (Slider)	(-0.75, -0.66, 0.00)	External lateral undercut with clear side-draw access along (-0.75, -0.66, 0.00) (-X); requires cam/hydraulic slide.
402	Side-Action (Slider)	(-1.00, -0.00, 0.00)	External lateral undercut with clear side-draw access along (-1.00, -0.00, 0.00) (-X); requires cam/hydraulic slide.

Face ID	Resolution	Action Vector	Engineering Rationale & Tooling Requirement
403	Side-Action (Slider)	(-0.83, -0.56, 0.00)	External lateral undercut with clear side-draw access along (-0.83, -0.56, 0.00) (-X); requires cam/hydraulic slide.

5. Parting Line & Shut-Off Quality

Loop Closure Status:	CLOSED LOOP (Valid)	Planarity & Shut-off:	Planar 2D parting line
Flashing Risk:	LOW	Loop Vertex Count:	5 points

Tooling Note: Planar parting line with uniform shut-off and minimal flash risk.

6. Mold Assembly & Tooling Split Volumes

Tooling Component	Volume (mm³)	Volume % of Stock Block	Ejection / Action Role
Master Stock Block	462080.01	100.0 %	Raw Tooling Steel/Aluminum Billet Envelope
Top Cavity Block (A-Side)	216101.37	46.8 %	Forms cosmetic class-A outer surface
Bottom Core Block (B-Side)	135080.29	29.2 %	Forms functional interior & houses ejector system
Side Slider (Left / -X)	47917.70	10.4 %	Lateral mechanical slide for undercut extraction
Side Slider (Right / +X)	47917.71	10.4 %	Lateral mechanical slide for undercut extraction

7. Extended DFM Analysis (Physics & Moldability)

7.1 Wall Thickness & Sink Mark Evaluation

Nominal Thickness: 11.45 mm | Min: 1.32 mm | Max: 28.74 mm | Variation: 239.5% (EXCEEDS ±15% TOLERANCE) | (Raycast Heuristic • Confidence: MEDIUM)

Warning: Found 10 isolated thick section(s) >1.3× nominal wall. Risk of sink marks, extended cooling time, and internal vacuum voids.

7.2 Injection Gate Placement Candidates

Candidate	Gate Type	Location (X, Y, Z)	Suitability	Design Rationale
Gate #1 (Face 17)	Pin-point / Direct Sprue Gate	(0.0, -0.0, 4.5)	94 / 100	Accessible planar exterior surface (Pin-point / Direct Sprue Gate) with direct mold flow path into nominal wall.
Gate #2 (Face 40)	Pin-point / Direct Sprue Gate	(-0.0, -0.0, 40.0)	91 / 100	Accessible planar exterior surface (Pin-point / Direct Sprue Gate) with direct mold flow path into nominal wall.
Gate #3 (Face 44)	Pin-point / Direct Sprue Gate	(-6.7, -8.0, 32.0)	90 / 100	Accessible planar exterior surface (Pin-point / Direct Sprue Gate) with direct mold flow path into nominal wall.

7.3 Rib/Boss Proportions, Corner Radii & Demolding

Rib / Boss Sizing:	Detected 149 cylindrical boss features. Base thickness target: 40-60% nominal wall (4.6-6.9mm).
Corner Radii / Stress:	Min measured fillet radius: 0.25mm (Target: $\geq 5.72\text{mm}$). Internal radii should be $\geq 5.72\text{ mm}$ (0.5x nominal wall) to mitigate notch stress and resin flow restriction.
Ejector Pin Layout:	Identified 8 stable core-side candidate locations away from thin walls.

8. Known Gaps & Explicit Limitations

In accordance with rigorous engineering practice, any physical phenomena uncomputable from static BRep CAD topology are transparently documented below rather than approximated:

Analysis Scope / Domain	Technical Reason & Derivation Limitation
Volumetric Warpage & Material Shrinkage PVT Compensation	Uniform nominal geometry analyzed without non-isotropic polymer PVT shrinkage curves for material 'PP'. Requires nonlinear finite-element injection molding warpage solver.
Cooling Circuit Conformal Optimization & Cycle Time Thermal Analysis	MHD/thermal transient CFD cooling simulation kernel is not integrated; cooling line optimization requires 3D transient heat flux modeling.
Weld Line Formation & Air Trap Location Prediction	Requires dynamic multi-phase cavity filling flow front simulation (Moldflow/OpenFOAM), uncomputable from static BRep topology alone.